The Kent Equation: A Time-Indexed Threshold Model for the Onset of Local Biological Self-Instantiation

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June 14, 2026

Abstract

We introduce the Kent Equation, a time-indexed threshold model that estimates the probability $K_{\text{tech}}(t)$ that a planetary civilization has crossed the technical threshold required to fabricate a viable biological body, to specification, on its own surface. The model is a cumulative distribution in $[0, 1]$ over a single crossing time, driven by a substrate-maturity index $M(t)$ — built from energy, general-purpose computation, and biological experimental throughput, all empirically trackable — and modulated by an artificial-intelligence accelerant that converts raw substrate capacity into solved scientific capability. We take the crossing time to be the later of two necessary, demonstrably separable technical gates, genome-scale synthesis and developmental viability, and show by explicit audit that the gates are distinct, that synthesis crosses earlier, and that developmental viability is therefore binding — which licenses a one-gate canonical model $K_{\text{tech}}(t) = F_{\text{dev}}(t)$. The framework is agent-neutral: *when can this world grow bodies?* is logically prior to, and separable from, any claim about who might occupy them. We instantiate at human (multicellular vertebrate) scale and develop the Bootstrap Hypothesis — that transport-limited extraterrestrial visitors would rationally wait for a host civilization to mature its biotechnology rather than transport bodies across interstellar distances — as one motivating application filling the model's premise slot; its waiting advantage follows from the locality of developmental knowledge, and the same mechanism applies unchanged to other embodiment questions, such as biological substrates for artificial agents. This is a conditional timing model, not an existence proof.

1 Introduction

At what point in a planetary civilization's technological development does it become possible to fabricate viable biological bodies, to specification, locally — to convert biological information into a living organism on the planet's own surface? We call a civilization that has crossed this threshold *instantiation-capable*, the threshold the *bootstrap point*, and we introduce the Kent Equation to estimate the probability $K_{\text{tech}}(t)$ that the bootstrap point has been reached on or before time t.

In factoring an intractable question into terms that different disciplines can attack separately, the model follows the example of the Drake Equation [1, 2, 3]; it shares that structural ambition while differing in subject and in form, replacing a static expected count with a time-indexed probability. That is the whole of its debt; the construction below stands on its own.

The model has four features we rely on throughout. First, it is a cumulative distribution function over a single threshold-crossing time, which lets it represent an approaching threshold and an

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accelerating rate of approach. Second, its driver — the substrate-maturity index $M(t)$ — is built from components (energy, general-purpose computation, biological experimental throughput) that have real, long-run, well-sourced time series, so the quantity driving the estimate can actually be tracked. Third, it separates raw substrate *capacity* ($M(t)$) from the *capability* to convert that capacity into solved scientific problems, modelling artificial intelligence as an accelerant on conversion rates rather than on capacity — a distinction validated by the historical record of computational biology. Fourth, its term set is audited down to a single binding gate by explicit construction: the threshold decomposes into two genuinely separable technical gates — genome-scale synthesis and developmental viability — of which synthesis crosses earlier, so developmental viability is binding and the canonical model collapses to $K_{\text{tech}}(t) = F_{\text{dev}}(t)$. The audit licenses this reduction rather than the model asserting it, and it discards no real variable, because separability establishes that the gates are distinct.

The framework is deliberately agent-neutral: the question *when can this world grow bodies?* is logically prior to, and separable from, any claim about who might wish to occupy them. We instantiate the model at human (multicellular vertebrate) scale and develop the Bootstrap Hypothesis — that transport-limited extraterrestrial visitors would rationally wait for a host civilization to mature its biotechnology rather than transport bodies across interstellar distances — as one motivating application that fills the model's premise slot. We show the waiting advantage follows from the locality of developmental knowledge, and the same mechanism applies without modification to other embodiment questions, such as the local instantiation of biological substrates for artificial agents. We are explicit throughout: this is a conditional timing model, not an existence proof.

2 Criteria for a legitimate factored estimator

A factored estimator is legitimate — independently of whether its numerical output is reliable — when it satisfies three structural properties, and we hold the model to them explicitly in Section 4.

1. **Valid factorization.** The terms must form a genuine decomposition, so that the chosen combining operation is mathematically correct. For a product of probabilities this requires the terms to be conditionally independent, or to be constructed as a chain of conditional probabilities $P(A)\,P(B \mid A)\,P(C \mid B)\cdots$. Terms that are positively correlated but multiplied as if independent systematically distort the product.

2. **Separate estimability.** Each term should, in principle, be estimable on its own, so that the decomposition does real work — isolating sub-questions and localizing disagreement.

3. **Dimensional coherence.** The units must resolve to the declared output type.

The relevant test is therefore not whether the estimator yields a good number, but whether it is a valid factorization of separately estimable terms with a coherent output.

3 The model

We construct the model in stages. We fix the output object (§3.1), introduce a general threshold as the last of several necessary crossings (§3.2), give a hazard formulation that routes all dependence through a shared driver (§3.3), build that driver $M(t)$ (§3.4), and locate intelligence as an accelerant on conversion rather than capacity (§3.5). We then state the canonical model (§3.6), whose single-gate form is justified, not assumed, by the audit of Section 4.

3.1 Output type and the agent-neutral object

We fix what $K_{\text{tech}}(t)$ is first, because the choice disciplines every term.

Definition 1 (Output)**.** Let the organism class Ω be a parameter of the model. $K_{\text{tech}}(t;\Omega)$ is the probability that a host civilization has crossed the technical threshold required to fabricate, to specification and on its own surface, a viable biological body of class Ω, on or before time t, conditional on the substrate trajectory $M_{[0,t]}$:

$$K_{\text{tech}}(t;\Omega) = P\big(\tau^* \le t \;\big|\; M_{[0,t]}\big)\,, \qquad K_{\text{tech}}(t;\Omega) \in [0,1], \tag{1}$$

where τ^* is the threshold-crossing time of §3.2.

For the worked model and for the Bootstrap application of Section 6 we instantiate Ω at human (multicellular vertebrate) scale; we write $K_{\text{tech}}(t)$ for that instance and carry Ω explicitly only where the organism class matters. Parametrizing rather than hardwiring the class fixes the well-definedness of the central random variable while preserving the agent-neutral generality of Section 7.

K_{tech} is a cumulative distribution function (CDF) over a single random variable τ^*. It is non-decreasing in t and rises toward 1 as the host's substrate matures — the property a static product cannot supply, and the reason the model is time-indexed.

Crucially, K_{tech} is agent-neutral: it asks only when a world can grow bodies, a question logically prior to and separable from who might wish to occupy them. The mechanism contains no application-premise term. An application is imposed instead by assuming its premise as background and reading K_{tech} under that assumption. Because τ^* depends only on the host's substrate and carries no agent-specific content,

$$P\big(\tau^* \le t \mid M_{[0,t]},\ \text{premise}\big) = P\big(\tau^* \le t \mid M_{[0,t]}\big) = K_{\text{tech}}(t), \tag{2}$$

so the premise contributes nothing to the dynamics. If one wants the application's quantity of interest to vanish when its premise fails, the relevant object is $\mathbb{1}[\text{premise}] \cdot K_{\text{tech}}(t)$, an outer bookkeeping factor that leaves the mechanism untouched. This is the formal content of Section 7.

3.2 The threshold as a maximum of crossing times

A natural but incorrect approach would multiply maturity scores for several capabilities. Because the maturities of a single developing civilization are strongly positively correlated, such a product violates the factorization criterion of Section 2: it double-penalizes correlated factors. We instead model the threshold as a conjunction of necessary crossings.

Definition 2 (Crossing times and the threshold)**.** For each necessary capability c in a candidate set $\mathcal{C}$, let τ_c be the first time capability c crosses the level required for biological instantiation of class Ω. The bootstrap point is reached when all required capabilities have crossed, so the threshold-crossing time is

$$\tau^* = \max_{c \in \mathcal{C}} \tau_c. \tag{3}$$

The maximum is correct precisely because the capabilities are necessary and conjunctive: readiness is gated by the last to arrive. This formulation is robust to the correlation that breaks the naive product; indeed it expects the underlying drivers to be correlated — a fast-developing civilization crosses several thresholds in close succession — and represents that correlation natively by routing every crossing time through a shared driver. The composition of $\mathcal{C}$ is not assumed here; it is audited in Section 4, where we establish that the candidate set reduces to a single binding gate.

3.3 Hazard formulation and the substrate driver

We model each crossing time through its hazard — the instantaneous rate of crossing given that crossing has not yet occurred — and we make every hazard a function of a common substrate-maturity trajectory $M(t)$. This is how the required dependence enters honestly: the gates are not independent, because they are all driven by $M(t)$, and the model says so.

$$P\big(\tau_c \leq t \mid M_{[0,t]}\big) = 1 - \exp\!\left(-\int_{t_0}^{t} h_c(M(s))\, ds\right), \tag{4}$$

where $h_c(\cdot)$ is the hazard for capability c, increasing in substrate maturity, and t_0 is a chosen epoch.

In the general multi-gate apparatus, if the crossings are conditionally independent given $M(t)$ — a far weaker and more defensible assumption than unconditional independence, since the common driver carries the bulk of the dependence — the joint threshold CDF factors as $\prod_{c\in\mathcal{C}}\big(1 - \exp(-\int h_c)\big)$. We retain this product only as scaffolding. The audit of Section 4 collapses $\mathcal{C}$ to a single gate, at which point the product disappears and no conditional-independence assumption survives in the canonical model.

3.4 The substrate-maturity index $M(t)$

The driver must be measurable, or the model gains formalism without empirical traction. We construct it from substrate components and subject membership to two rules.

Assumption 1 (Exclusion, re-scoped)**.** No component of $M(t)$ may itself be a gated capability. The gated capability is developmental viability (τ_{dev}, Definition 4); it is excluded from $M(t)$, since it is precisely the quantity $M(t)$ is meant to predict. Including it would make the model circular.

The scope of this exclusion is narrower than "exclude all biotechnology," and the narrowing is load-bearing. Three capacities sit upstream of the gate and are admitted: sequence design (computing which sequence to write — a computational capability, exercised long before gestation), and biological experimental throughput (the capacity to run developmental experiments: automated embryo culture, organoids, imaging, closed-loop wet-lab automation), which is the capacity layer for development as distinct from achieved viability, in the same capacity-versus-capability sense used throughout. Physical genome-synthesis throughput governs the non-binding synthesis gate (Definition 3) rather than the developmental hazard, so it too is kept out of $M(t)$ and monitored separately as a prerequisite indicator. Only achieved developmental viability — the binding gate — is excluded, which resolves rather than asserts away the circularity a developmentally-relevant proxy would otherwise raise.

Assumption 2 (Mechanistic membership)**.** A component is admitted to $M(t)$ only if it has a named causal pathway to the gated capability, not merely a correlated upward trend.

This filter is essential because, in any era of broad technological growth, nearly every index rises together; co-trending alone is close to uninformative. Empirical content must be sought in the ordering and timing of crossings, not in the shared direction of growth. Under these rules we adopt a minimal three-component index; because the index drives the developmental hazard specifically, its components are chosen for their pathway to developmental viability.

Component	Representative measurable proxy	Named pathway to τ_{dev}
Energy (E)	Generation capacity; real cost per unit energy	Upstream of all computation and experimentation
Computation (C)	Cost per FLOP; installed capacity	Carries the inference that performs sequence design and developmental simulation
Biological experimental throughput (B)	Automated embryo-culture / organoid / imaging / wet-lab capacity	Generates the empirical developmental knowledge that viability requires

These three are causally stacked — energy upstream of computation, computation upstream of experimental throughput — which aids coherence but does not by itself defeat co-trending; if anything, a dependency chain produces more lockstep trajectories. What defeats co-trending is mechanistic membership plus the retreat of empirical content to timing and ordering. Stacking also makes the components partially redundant, so $M(t)$ has fewer effective degrees of freedom than its three inputs suggest.

Aggregation. The components are combined by a weighted geometric mean rather than a weighted sum. The choice is substantive: a sum treats components as substitutable, whereas the substrate is complementary — all layers are required, and a near-absence of any one should drive the index toward zero. A civilization with abundant energy and experimental throughput but negligible general-purpose computation (a "steampunk" world) cannot perform the design and developmental inference on which fabrication depends; the geometric mean correctly assigns it near-zero maturity, whereas a weighted sum would wrongly score it as moderately mature.

Normalization. Components cannot be combined in native units and have no natural maximum, since they are open-ended growth quantities. Normalizing to a present-day maximum would force $M(t) \approx 1$ today by construction and forbid further maturation — fatal for a time-indexed model. We therefore normalize each component to the level it must reach for the gate, denoted x_i^*:

$$M(t) = \prod_{i \in \{E,C,B\}} \left(\frac{x_i(t)}{x_i^*} \right)^{w_i}, \qquad \frac{x_i(t)}{x_i^*} \leq 1 \text{ (capped per component)}, \tag{5}$$

with weights w_i and a per-component cap that prevents a single racing component from masking a lagging one. This relocates the model's principal free assumption from an arbitrary scaling maximum to the required levels x_i^* — a relocation we regard as an improvement, since the required levels are a substantive scientific question on which domain experts can disagree productively. We do not claim the required levels are known; we claim only that this is the right place for the uncertainty to live. With a single binding gate (§3.6), the x_i^* are unambiguously the levels required for developmental viability.

3.5 The intelligence accelerant

The components of $M(t)$ are capacity variables: they measure how much raw substrate exists. They do not measure how effectively that capacity is converted into solved scientific problems. The conversion is a distinct capability, and it is the natural locus of artificial intelligence.

The distinction is not merely conceptual. The historical record of computational biology shows raw computational capacity present for years before the capability to use it on specific biological problems existed — structure prediction being the paradigmatic case [6, 7], where the bottleneck that broke was one of inference, not of available computation. This is exactly the signature of a capability separate from capacity.

We therefore model intelligence as an accelerant on the conversion rate, not as a component of capacity. Writing $h_c(\cdot)$ for the baseline (brute-force) hazard as a function of substrate and $\mathcal{A}_c(t)$ for the accelerant on gate c, the effective hazard is

$$\tilde{h}_c(t) = \mathcal{A}_c(t) \cdot h_c(M(t))\,. \tag{6}$$

Proposition 1 (Baseline, and the absence of a floor)**.** *The accelerant satisfies $\mathcal{A}_c(t) > 0$, with $\mathcal{A}_c = 1$ defined as the no-net-acceleration baseline: the epoch in which substrate is converted at its brute-force, capability-neutral rate. Values $\mathcal{A}_c > 1$ represent net acceleration; values $0 < \mathcal{A}_c < 1$ represent net drag — institutional, epistemic, safety, or misuse-driven friction that converts substrate less efficiently than the brute-force baseline. We do not impose $\mathcal{A}_c \geq 1$: an accelerant measured relative to a chosen baseline can fall below that baseline, and real research systems exhibit such drag.*

Remark (Restricted-AI civilizations are slowed, not blocked). The guarantee that a civilization wary of AI still reaches the threshold does not rest on a floor under the accelerant; it rests on the positivity of the baseline hazard. A gate is capacity-crossable when $h_c(M(t)) > 0$ for mature $M(t)$. Since developmental viability is capacity-crossable (§4.2), even an epoch of net drag ($\mathcal{A}_{\text{dev}} < 1$) has effective hazard $\mathcal{A}_{\text{dev}} \cdot h_{\text{dev}}(M) > 0$, so the threshold is still crossed — only later. Acceleration and drag set the speed of crossing; positivity of the baseline hazard guarantees that crossing occurs.

A uniform accelerant acting equally on all gates would be a separate modelling commitment, but the single-gate reduction of §3.6 makes the gate-specific-versus-uniform distinction vacuous in the canonical model: there is one gate and one accelerant, $\mathcal{A}_{\text{dev}}$. The comparative accelerant profile across the two candidate gates — synthesis heavily accelerated, development only modestly — is invoked in §4.2, where it establishes the crossing order.

3.6 The canonical (one-gate) model

The term audit of Section 4 establishes two facts about the candidate set $\mathcal{C} = \{G, \text{dev}\}$: the gates are genuinely separable (both "write-but-can't-grow" and "grow-but-can't-write" worlds are constructible, so neither term is redundant), and on the realized technological trajectory synthesis crosses first, so development is the binding gate. The threshold therefore collapses, $\tau^* = \max(\tau_G, \tau_{\text{dev}}) = \tau_{\text{dev}}$, and the canonical Kent Equation is the single-gate CDF:

$$K_{\text{tech}}(t) = F_{\text{dev}}\big(t \mid M_{[0,t]}\big) = 1 - \exp\left(-\int_{t_0}^{t} \mathcal{A}_{\text{dev}}(s)\, h_{\text{dev}}(M(s))\; ds\right), \tag{7}$$

with $M(t)$ the single substrate index of §3.4, $\mathcal{A}_{\text{dev}}$ the two-sided developmental accelerant of §3.5, and physical genome-synthesis throughput a necessary prerequisite that crosses earlier and is monitored separately but does not appear in the binding hazard.

The reduction discards a product and with it any conditional-independence assumption; it leaves a single substrate index rather than a vector or gate-indexed family, and a single, two-sided accelerant rather than a constrained set. The general multi-gate apparatus of §§3.2–3.3 is retained only as the frame within which Section 4 demonstrates the reduction; the object the rest of the paper uses is the one-gate F_{dev}. The canonical object is premise-free: the Bootstrap application conditions on its premise as background, which leaves K_{tech} unchanged and merely makes it the quantity worth watching.

4 Term audit

We now hold the model to the factorization and separate-estimability criteria of Section 2 by auditing the candidate gate set $\mathcal{C}$. The audit does two distinct jobs, answered with the same evidence but logically separate:

1. **Separability — a claim about logical possibility.** Are the candidate gates genuinely distinct variables, or is one a relabelling of the other? We test this by world construction: can a world be built in which one gate has crossed and the other has not, in both directions? If both directions are constructible, the gates are distinct and neither is a hidden redundancy.
2. **Ordering — a claim about the realized trajectory.** Given that the gates are distinct, which one crosses last? Because $\tau^* = \max_c \tau_c$, the last gate to cross is the binding constraint; if one gate reliably crosses first along the trajectory civilizations actually follow, the maximum collapses onto the other.

Separability and binding are not in tension: a gate can be logically separable while being non-binding on the realized path. The audit therefore (i) proves the gates are real and distinct, so the collapse discards no information, and (ii) identifies which gate dominates the clock.

Definition 3 (Genome-synthesis gate)**.** τ_G is the time at which it becomes reliably possible to physically synthesize and fidelity-validate a specified genome — to write a correct, error-checked sequence faithful to its intended design. Its output is correct biological information.

Definition 4 (Developmental-viability gate)**.** τ_{dev} is the time at which a specified genome can be grown into a viable, supported organism of class Ω — gestation or its technological equivalent, together with the developmental and immunological support required to proceed from sequence to living body. Its output is a body from a sequence.

The distinction in one line: τ_G yields correct information; τ_{dev} turns that information into a body. Writing and growing are different operations.

The placement of design (the narrow carve). A civilization must also design the sequence. We split design along its two sub-problems, and the split is load-bearing for the one-gate model. Design-for-synthesizability — computing a writable sequence that realizes an intended specification — is a computational inference capability; it lives in $M(t)$ (§3.4), not in any gate. Design-for-viability — determining which specification actually yields a viable organism, for a never-before-existing target such as an environment-adapted hybrid — cannot be settled by inference alone, because viability for a novel organism is confirmed only by running development; it therefore routes through τ_{dev}. Under this carve, τ_{dev} absorbs two things — can we grow the body? and do we know which sequence grows? — and τ_G is narrowed to physical writing with fidelity. This is a deliberate choice with a plain cost — τ_{dev} does substantial work — adopted because it matches where the empirical bottleneck sits (§4.2) and keeps the agent-neutral separation clean: who supplies the design is premise content, while design itself is a computational capability in $M(t)$. "Design is not a separate gate" is not "design is unmodeled."

4.1 Separability of the two technical gates

We construct both directions.

Can τ_G cross without τ_{dev}? Yes — and this direction is close to actual. Genome writing has already reached genome scale across viruses, bacteria, and now a single-celled eukaryote: the synthetic-yeast effort has written chromosomes of *Saccharomyces cerevisiae* de novo [5], building on the first chemically synthesized bacterial genome a decade earlier [4]. Yet no technology can grow a specified multicellular vertebrate body from such a sequence (§4.2). The capacity to write a long, correct sequence confers no capacity to grow it; developmental support is a separate technical stack that lags synthesis badly. This direction is not merely constructible — it is roughly where the technology stands.

Can τ_{dev} cross without τ_G? Yes, though the world is more exotic. A civilization with mastery of gestation and developmental support, applied to natural or copied genomes but lacking reliable de novo synthesis, could instantiate bodies from existing genomes while unable to author novel ones. Gestating an existing genome and authoring a new one are different problems; the former does not require the latter.

Both directions are constructible, so τ_G and τ_{dev} are genuinely separable; neither is redundant. This refutes the natural suspicion that synthesis and development are the same variable because validation appears developmental: fidelity-validation (is the written information correct relative to the design?) is an information-checking operation belonging to τ_G, whereas viability (will the design grow into a body?) is an instantiation operation belonging to τ_{dev}.

4.2 Accelerant types, crossing order, and the binding constraint

Both technical gates are capacity-crossable — $h_c(M(t)) > 0$ for mature $M(t)$ — but they differ sharply in how much intelligence accelerates them, and therefore in when they cross.

Synthesis (τ_G) is substantially a problem of throughput, error correction, and cost, and is in addition the gate that intelligence most dramatically accelerates: sequence and structure design are paradigmatic inference problems, the class on which AI has already produced step-changes [6, 7]. It has a high baseline hazard and a large accelerant, and crosses early.

Development (τ_{dev}) is gated less by computational throughput than by empirical biological knowledge — morphogenesis, immunological integration, developmental signaling — much of which must be obtained by running real developmental processes rather than by simulation alone. Inference can design, but it cannot observe a process that has not been run. Development therefore has a lower, less capacity-responsive hazard and a modest accelerant, and crosses late.

The current empirical record supports this ordering directly, and at human scale the asymmetry is extreme (Table 1). The genome-writing evidence is overwhelmingly microbial and unicellular — strong evidence that writing scales with throughput and cost, weak evidence for τ_G at vertebrate scale, which strengthens rather than undercuts the binding claim, since it shows synthesis advancing precisely along the axis that capacity and intelligence accelerate. The developmental frontier, by contrast, reaches only the two ends of gestation, with continuous conception-to-birth instantiation of a specified body entirely absent and no path that simulation alone could supply.

Proposition 2 (Development is the binding constraint — argued, on present trajectories)**.** *Given the current state of synthetic genomics, embryo culture, and artificial-womb technology, developmental viability is the plausibly binding gate: along the trajectory a civilization actually follows, $\tau_G \le \tau_{\text{dev}}$, so $\tau^* = \max(\tau_G, \tau_{\text{dev}}) = \tau_{\text{dev}}$. Synthesis crosses early under heavy acceleration; development, empirically bottlenecked and only modestly accelerated, is the last gate to cross. We state this as an argued claim and modelling premise, not a proved proposition: it rests on the evidence above and on the accelerant-profile argument, either of which future developments could shift.*

The human-scale sharpening. The single-gate collapse of §3.6 is exact when $\tau_G \le \tau_{\text{dev}}$ with probability one, and a good approximation when that ordering holds with high probability; the residual it discards is the probability that development has crossed while synthesis has not,

$$K_{\text{tech}}(t) = F_{\text{dev}}(t) - \underbrace{P(\tau_{\text{dev}} \le t,\ \tau_G > t)}_{\text{“grow-but-can't-write” world}} \quad . \tag{8}$$

Table 1: The synthesis/development asymmetry at the current frontier. Genome writing has reached single-celled-eukaryote scale along the throughput/cost axis that capacity and intelligence accelerate; developmental viability addresses only the two ends of gestation, with no demonstrated path to continuous instantiation of a specified body.

	Genome synthesis (τ_G)	**Developmental viability (τ_{dev})**
Frontier demonstrated	Viral, bacterial [4], and single-celled eukaryotic genomes written de novo [5]	Mouse embryos cultured ex utero from pre-gastrulation to late organogenesis [8]; stem-cell embryo models reach early organogenesis
Scale reached	$\sim$12 Mb (yeast, 16 chromosomes)	A few days of a $\sim$20-day mouse gestation; human embryo culture limited to $\sim$14 days
Human-scale status	Not yet demonstrated (human genome $\sim$3 Gb), but the obstacle is throughput, cost, and fidelity, which capacity and AI directly attack	Synthetic human embryo models form placenta and yolk sac but not a beating heart or brain; artificial-womb technology supports only the last gestational third [9, 10], not conception-to-birth
Rate-limiter	Inference + synthesis throughput — strongly AI-accelerable	Empirical developmental knowledge, obtainable only by running development — weakly AI-accelerable

Instantiating at human scale makes this correction term negligible. At vertebrate scale a world that can gestate a specified, novel body but cannot write its sequence runs against the actual direction of progress on both gates: writing is the capability that is advancing, and growing is the capability that is far behind. The "grow-but-can't-write" world remains logically constructible (§4.1) — which is why synthesis stays in $\mathcal{C}$ as a genuine variable — but its probability mass on the realized human-scale trajectory is approximately zero. Human-scale instantiation is therefore exactly the regime in which development binds strictly enough that the collapse to F_{dev} is an identity rather than a mere convenience.

4.3 The social term is not a gate

A third term is commonly conflated with the technical gates: the capacity of a society to absorb the local instantiation of non-human biological persons. Call its crossing time τ_S. We first sharpen it, because in its loose form — the situation is introduced, hidden, governed, normalized, or otherwise managed — it is satisfied in every possible world and is therefore not a variable but the constant 1. A term true in all worlds fails the estimability criterion and must be sharpened or removed.

Definition 5 (Sharpened social term)**.** τ_S is the time at which the social, legal, and legitimacy substrate can absorb local instantiation without systemic rejection. Concealment does not count as crossing: a perpetually hidden program leaves τ_S uncrossed, since nothing has been absorbed. A world of stable, self-reinforcing, indefinite prohibition has $\tau_S = \infty$.

Under this sharpening τ_S is falsifiable. Two observations then remove it from the gating set $\mathcal{C}$. First, τ_S is largely orthogonal to the substrate $M(t)$: social absorption capacity is not manufactured by energy, computation, or biological experimental throughput, so the hazard h_S is approximately independent of $M(t)$. We take intelligence to be approximately neutral on it as a baseline ($\mathcal{A}_S \approx 1$), though the two-sided accelerant of §3.5 now lets the regime of AI-driven disruption that reduces absorption capacity and raises τ_S ($\mathcal{A}_S < 1$) be represented. Second, and decisively, τ_S is not a precondition for instantiation to be possible. Whether a society will calmly absorb instantiated non-human persons does not determine whether the host civilization *can* fabricate bodies; it determines only how the resulting situation unfolds. The social term therefore belongs not in the gating conjunction but in a downstream consequence layer: it governs the observable social dynamics that accompany the approach of the technical threshold, not whether the threshold is reached.

Proposition 3 (Demotion, not deletion). *τ_S is removed from the gating set, giving $\mathcal{C} = \{G, \mathrm{dev}\}$ and $\tau^* = \max(\tau_G, \tau_{\mathrm{dev}}) = \tau_{\mathrm{dev}}$, and the social variable is retained as a consequence layer downstream of K_{tech}, describing the social phenomena expected as K_{tech} rises. It is a function of how far the technical clock has run, not a term in the clock.*

4.4 The audited equation

Collecting the audit results: separability (§4.1) establishes that τ_G and τ_{dev} are genuinely distinct, so neither can be dropped as redundant; the crossing-order argument and its evidence (§4.2) establish that synthesis crosses first, so development is binding; and the demotion (§4.3) removes the social term. The gating set reduces to a single binding gate, and the canonical Kent Equation is the single-gate CDF (7), with $\tau^* = \tau_{\mathrm{dev}}$ and genome synthesis retained as a necessary, demonstrably earlier, non-binding prerequisite.

The factorization criterion of Section 2 is now met by a single-term factorization — there is no product, so there is no independence assumption to defend. Each remaining quantity is separately estimable in principle, and the output is a coherent probability in $[0, 1]$. The model meets the criteria with the smallest term set the audit permits.

5 Illustrative sensitivity

The contribution of this paper is a structure, not a fitted set of values (Section 8), and we do not calibrate it. But a structure should be shown to behave as claimed. We exhibit a deliberately minimal parametrization and confirm three properties asserted above: that the clock accelerates, that the accelerant is two-sided but non-blocking, and that the crossing order licenses the one-gate collapse. None of the values below is empirical.

We let $M(t)$ rise smoothly toward its required level along a logistic ramp, and route both candidate gates through it via baseline hazards that differ in their responsiveness to substrate, $h_{\mathrm{dev}}(M) = k_{\mathrm{dev}} M^3$ and $h_G(M) = k_G M^{1.2}$, with each crossing-time CDF given by $F_c(t) = 1 - \exp(-\int_0^t \mathcal{A}_c h_c(M(s))\, ds)$ and $K_{\mathrm{tech}}(t) = F_{\mathrm{dev}}(t)$. Development's high exponent encodes that it is less capacity-responsive — its hazard stays near zero until substrate is well matured — while synthesis's low exponent and larger coefficient encode that it is strongly capacity-responsive, rising early. The accelerant enters multiplicatively and is two-sided (§3.5).

Figure 1 confirms all three. Panel (a) shows the CDF non-decreasing and steepening, and the family ordered by $\mathcal{A}_{\mathrm{dev}}$ with drag and acceleration on one axis, the lowest curve still reaching 1 — behaviour no static estimator can have. Panel (b) shows the binding argument of §4.2 produced rather than assumed: given only that synthesis is more capacity-responsive and more heavily accelerated than development, the model places τ_G before τ_{dev} and drives the discarded mass to near zero on its own. The figure shows the model *can* produce an accelerating clock with a binding developmental gate under plausibly-shaped inputs; it does not show when the real threshold is crossed — that requires estimating x_i^*, the hazard forms, and the accelerant magnitudes, the empirical program left to subsequent work. With one gate the model has few knobs — a single

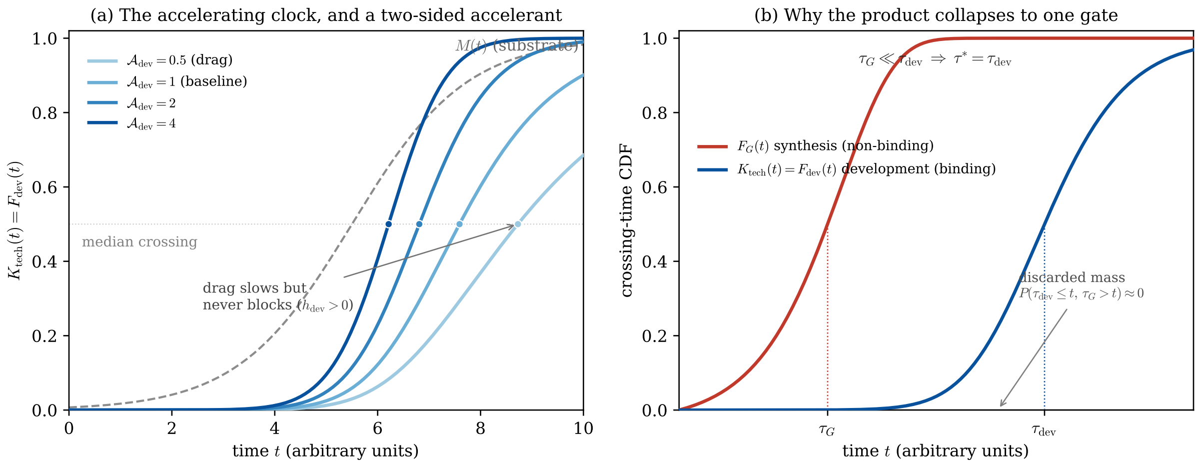

Figure 1: Illustrative behaviour of the one-gate Kent Equation under a toy substrate ramp $M(t)$ (dashed, panel a). Values are arbitrary; the figure demonstrates qualitative behaviour, not a calibration. **(a)** The canonical $K_{\text{tech}}(t) = F_{\text{dev}}(t)$ for developmental accelerants $\mathcal{A}_{\text{dev}} \in \{0.5, 1, 2, 4\}$. Each curve steepens as $M(t)$ ramps — the accelerating clock — and stronger acceleration shifts the median crossing earlier (from $t \approx 8.7$ under drag to $t \approx 6.2$ under heavy acceleration). The drag case $\mathcal{A}_{\text{dev}} = 0.5$ crosses latest but still crosses: with $h_{\text{dev}}(M) > 0$, a restricted-intelligence civilization is slowed, never blocked. **(b)** Under a shared substrate, synthesis ($\mathcal{A}_G = 4$) crosses early at $\tau_G \approx 2.9$ while development ($\mathcal{A}_{\text{dev}} = 1.5$) crosses late at $\tau_{\text{dev}} \approx 7.1$. Because $\tau_G \ll \tau_{\text{dev}}$, $\tau^* = \tau_{\text{dev}}$ and $K_{\text{tech}} = F_{\text{dev}}$. The discarded "grow-but-can't-write" mass $P(\tau_{\text{dev}} \leq t,\ \tau_G > t)$ (grey) peaks at ≈ 0.001 — the quantitative form of the §4.2 claim that the one-gate collapse is an identity in this regime.

index, hazard, and two-sided accelerant — the analytic dividend of the reduction: a vector-valued formulation would require sweeping a high-dimensional space and invite the over-fitting objection the model is built to avoid.

6 Application: the Bootstrap Hypothesis

We now fill the premise slot of the canonical equation with the application that motivates the model. We present the Bootstrap Hypothesis as a structured proposal and are explicit that it is speculative: nothing in Sections 3–5 depends on its truth, and we adopt the evidentiary posture of the field. The NASA Unidentified Anomalous Phenomena Independent Study Team found no evidence that reported UAP are extraterrestrial in origin, stating that in the peer-reviewed scientific literature there is to date no conclusive evidence suggesting such an origin, and treating extraterrestrial explanation as the hypothesis of last resort [11]. We do not claim the hypothesis's premise is true; we claim only that if its premise holds, the canonical equation is the instrument that times the consequence.

6.1 Statement

The Bootstrap Hypothesis supposes that extraterrestrial non-human intelligence has reached Earth, but that *reaching* a planet and *colonizing* it are governed by very different constraints. Interstellar transport of large biological populations, ecosystems, and industrial infrastructure may be subject to hard limits — on mass, volume, energy scaling, or biological payload — even for a civilization able to send small craft, observers, or samples. Under such limits, the rational colonization strategy is not to transport bodies across interstellar distances. It is to wait until the host planet has independently developed the technological substrate required to instantiate bodies locally, converting colonization from a transport problem into a local fabrication problem. The premise is the conjunction:

Term	Meaning
V	Visitation: the visitors can reach the host planet at least occasionally.
L	Limitation: large-scale biological transport is constrained.
W	Waiting advantage: delayed local instantiation is preferable to transported colonization.

The choice to instantiate at human scale enters the application directly. A visiting civilization is, by hypothesis, not adapted to Earth's biochemistry, atmosphere, microbial environment, or gravity; the probability that its unmodified genome yields an Earth-viable organism is vanishingly small. The visitors therefore require not merely a body but a designed hybrid — a genome authored to be viable in Earth conditions. This is why the application engages both technical capabilities of Section 4: the hybrid must be designed and written, and it must be grown. Given $V \wedge L \wedge W$, the quantity that matters is the host's progress toward the bootstrap point — what K_{tech} measures — and the conditioning structure of §3.1 applies: the premise is assumed as background, not multiplied into the mechanism, and because τ_{dev} depends only on the host's substrate $M(t)$, conditioning on $V \wedge L \wedge W$ leaves K_{tech} unchanged. The premise selects whether the clock is worth watching; it does not alter the clock's reading.

6.2 The waiting advantage is grounded in knowledge locality

The hypothesis invites an obvious objection. If the visitors are advanced enough to cross interstellar distances, why would they wait on the maturation of a less advanced host's biotechnology? Stated baldly, W looks like special pleading.

The interlocking of human-scale instantiation, the developmental-binding result of Section 4, and the agent-neutrality of Section 7 supplies a non-question-begging answer. The binding constraint is Earth-specific developmental knowledge, and advancement elsewhere does not purchase it. The argument runs in five steps.

1. **The visitors need an Earth-adapted hybrid:** not being Earth-adapted, their unmodified genome is not Earth-viable, so a designed hybrid is required.

2. **Designing the hybrid requires knowing which specification is Earth-viable.** This is not the synthesizability of the sequence — a computational question the visitors may well have solved — but its viability: whether a given design will develop into a functioning organism in Earth's environment.

3. **Earth-viability is Earth-specific knowledge.** It depends on Earth biochemistry, developmental signaling, immunological context, and gestational environment. It is not derivable from physics, and not transferable from the visitors' own biology; it is a fact about Earth developmental processes.

4. **Earth-specific viability knowledge accrues only by running Earth development.** This is the developmental bottleneck of §4.2 restated for a novel organism: inference can design, but it cannot observe a developmental process that has not been run. For a never-before-existing hybrid, viability is confirmed only empirically.

5. **Therefore no degree of computational advancement pre-solves the problem.** The visitors face the same τ_{dev} the host faces, because τ_{dev} is a property of running Earth development, not of who runs it.

This dissolves the objection without special pleading: the visitors' advancement is in the wrong currency. Their compute may design synthesizable sequences and their technology may write them — plausibly en route, long before arrival — but the two operations that require Earth-developmental capability, viability-design and growing, both route through τ_{dev}. The visitors then have two paths: independently rebuild Earth-developmental biology at scale, covertly and without the host's infrastructure — which is simply to become a covert instantiation-capable presence, the same mechanism performed by a different agent — or wait for the host to mature τ_{dev} and make use of the result. That the second is cheaper than the first is the content of W. The waiting advantage is not assumed; it follows from the locality of developmental knowledge. We are careful about scope: this strengthens the hypothesis's internal coherence, not its truth; the premise V remains assumed.

6.3 What the model contributes to the hypothesis

Feeding $V \wedge L \wedge W$ into the canonical equation has three consequences. Two follow from results already established. First, by the rising developmental hazard (§3.5, Figure 1), the relevant countdown is the rise of $K_{\text{tech}} = F_{\text{dev}}$, which *accelerates* as $M(t)$ matures — so the pressure the hypothesis associates with the approach is driven by a technical trajectory, not by social readiness. Second, by Proposition 3, the social dimension is a downstream consequence layer, so as K_{tech} rises the model predicts intensifying phenomena there — contestation over biotechnology, regulatory conflict, public pressure — with causation running from technical approach to social intensification, not the reverse (and, via the two-sided accelerant, admitting regimes of intelligence-driven social drag). The third is specific enough to state as a proposition.

Proposition 4 (Development is what to watch, concretely)**.** *With the one-gate reduction, developmental viability is not merely the binding gate but the only gate; the model directs all of an observer's attention to a single, monitorable class of technology. The evidence base of §4.2 supplies the concrete indicators: the duration over which embryos can be cultured ex utero; the completeness of synthetic embryo models — whether they advance from forming placenta and yolk sac to forming hearts and brains; and the gestational coverage of artificial-womb technology — whether it extends earlier than the final gestational third. Genome synthesis, which crosses earlier and is not binding, is explicitly not the signal; an observer tracking synthesis milestones would systematically misjudge proximity.*

These are conditional predictions: they describe what follows if $V \wedge L \wedge W$ holds. The Kent Equation supplies the mechanism; the Bootstrap Hypothesis supplies the premise and bears the speculative weight. A reader who rejects the premise loses the predictions but not the mechanism.

7 The agent-neutral generalization

The separation enforced in §3.1 — all application-specific content held in the background premise, all dynamics in the mechanism — has a consequence beyond tidiness: the mechanism is reusable. The question it answers, *when can this world fabricate viable bodies to specification?*, is logically prior to and independent of who might wish to occupy those bodies.

Proposition 5 (Agent-neutrality)**.** *The mechanism — the single developmental gate with its substrate driver and accelerant, $K_{\text{tech}}(t) = F_{\text{dev}}(t \mid M_{[0,t]})$ — contains no agent-specific content. The identity and motives of any party interested in the fabricated bodies enter only through the background premise of §3.1, which leaves K_{tech} unchanged. Swapping the application therefore swaps the* *premise while leaving the mechanism untouched. Even the audit that licensed the one-gate reduction (Section 4) is agent-neutral: separability and crossing-order are properties of the technology, not of any agent, so every application inherits the same single-gate model.*

The placement of design secures the generalization. The narrow carve of Section 4 — design-for-synthesizability located in $M(t)$, who supplies the design treated as premise content — is what makes the mechanism portable, and the reason is visible only when the application is swapped. It would be tempting, in the Bootstrap application, to model design as an external input the visitors provide; but that choice would not survive a change of agent, since an artificial agent seeking a biological substrate has no design apparatus of its own and may depend on the host's computation to design the sequence and the host's synthesis to write it. Locating design in the substrate rather than in the premise is exactly what keeps the mechanism agent-neutral.

What is invariant across all applications is τ_{dev}. This is the content of §6.2 generalized: the developmental bottleneck binds not because of any feature of the visitors, but because it is a property of running Earth development, which is indifferent to the agent who benefits.

Illustration: an artificial agent. Consider an artificial agent seeking biological embodiment. Such an agent faces no transport problem and has no visitation premise; its premise slot is filled differently — for instance: the agent exists, the agent seeks biological embodiment, local fabrication is its viable path. Yet the technical clock it depends on is identical: the same τ_{dev}, the same substrate driver $M(t)$, the same developmental bottleneck. The one operational difference — the agent may rely on the host's design and synthesis rather than supplying its own — lives entirely in the premise. Across the Bootstrap and artificial-agent applications the mathematics is untouched; what might appear to be a weakness of the model in its Bootstrap application, that the extraterrestrial content does no computational work, is in fact the source of this generality. The quarantining of agent-specific content into an assumed premise is exactly what makes the mechanism portable across embodiment questions.

8 Limitations and open problems

We catalogue the model's principal limitations and the points a careful referee should press.

Co-trending is mitigated, not eliminated. In an era of broad technological growth almost every index rises together, so correlation between $M(t)$ and the gated capability is nearly free and nearly uninformative; the model's empirical content lives in its timing and ordering claims, not in the shared upward direction. We regard this as a genuine constraint on testability.

The required levels x_i^* are unknown. The model relocates its principal free assumption to the substrate levels required for the developmental gate — a better home for the uncertainty than an arbitrary scaling maximum, but not its elimination. Estimating x_i^* is a substantive open problem in developmental biology.

The binding claim is an argued thesis, not a theorem. Proposition 2 rests on the current empirical asymmetry and on the accelerant-profile argument; a large, unanticipated advance in developmental simulation or de-novo viability prediction would weaken it, and with it the exactness of the one-gate collapse.

The narrow carve is a modelling choice. Placing viability-design inside τ_{dev} (Section 4) is what makes the reduction near-exact; housing it inside synthesis instead would yield gates that overlap more in time and a larger correction term (8), for which the general joint form, not the collapse, would be the honest object.

The social consequence layer is left qualitative. We decline to formalize the social variable as a function of K_{tech}, since doing so risks reintroducing the soft variable the audit removed; a quantitative consequence layer is a possible extension.

The model is a structure, not a calibration. The contribution is the equation and its decomposition, not a fitted set of values; any specific quantities one substitutes are illustrative (Section 5). We have specified what must be measured and where each measurement enters, and leave the empirical program to subsequent work.

9 Ethics, biosecurity, and dual-use considerations

A framework that models when a civilization becomes capable of fabricating arbitrary specified organisms locally has an evident dual-use dimension. We distinguish a forecasting instrument, which provides no enabling technical detail, from capability-enabling work, and locate this model firmly in the former category: it estimates *when* a threshold is approached from publicly trackable indicators, not *how* to cross it. A complete treatment — situating the model within synthetic-genome, developmental-biology, and artificial-womb governance and the biosecurity literature, and stating a responsible-disclosure posture for the developmental-viability indicators identified in §4.2 and Proposition 4 — is deferred to a subsequent revision.

10 Conclusion

The Kent Equation is a time-indexed, conditional estimator for the onset of local biological self-instantiation. Its driver is measurable; it separates substrate capacity from conversion capability and locates intelligence as a two-sided accelerant on the latter; and its term set is audited down to a single binding gate — developmental viability — with genome synthesis a separable but earlier, non-binding prerequisite. The framework is agent-neutral, with all application-specific content quarantined in a swappable premise, which both allows the modelling contribution to stand independently of any speculative application and makes the mechanism reusable across embodiment questions.

What the equation computes is the timing of a host-civilization technical capability: the probability that the threshold for local instantiation has been crossed by time t. It does not, by itself, establish anything about extraterrestrial intelligence, artificial agents, or any other party that might make use of that capability; all such content resides in an assumed premise that contributes nothing to the computation. We have presented the Bootstrap Hypothesis as the motivating application and shown that its waiting advantage follows from the locality of developmental knowledge, while being explicit that the equation does not establish the hypothesis: it supplies a clock, and the hypothesis supplies the premise that the clock is worth watching. Whether or not one finds the application persuasive, the mechanism is, we hope, a well-formed and honestly bounded instrument.

References

[1] F. D. Drake, "The radio search for intelligent extraterrestrial life," in G. Mamikunian and M. H. Briggs (eds.), *Current Aspects of Exobiology*, Pergamon Press, 1965.

[2] C. Sagan and I. S. Shklovskii, *Intelligent Life in the Universe*, Holden-Day, 1966.

[3] D. A. Vakoch and M. F. Dowd (eds.), *The Drake Equation: Estimating the Prevalence of Extraterrestrial Life Through the Ages*, Cambridge University Press, 2015.

[4] D. G. Gibson et al., "Creation of a bacterial cell controlled by a chemically synthesized genome," *Science* **329** (2010) 52–56.

[5] N. Annaluru et al., “Total synthesis of a functional designer eukaryotic chromosome,” *Science* **344** (2014) 55–58.

[6] J. Jumper et al., “Highly accurate protein structure prediction with AlphaFold,” *Nature* **596** (2021) 583–589.

[7] J. Abramson et al., “Accurate structure prediction of biomolecular interactions with AlphaFold 3,” *Nature* **630** (2024) 493–500.

[8] A. Aguilera-Castejón et al., “Ex utero mouse embryogenesis from pre-gastrulation to late organogenesis,” *Nature* **593** (2021) 119–124.

[9] E. A. Partridge et al., “An extra-uterine system to physiologically support the extreme premature lamb,” *Nature Communications* **8** (2017) 15112.

[10] U.S. Food and Drug Administration, Pediatric Advisory Committee, meeting on artificial womb technology for extremely premature infants, September 2023.

[11] NASA Unidentified Anomalous Phenomena Independent Study Team, *Final Report*, September 14, 2023.